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Career Readiness Report

A point-in-time assessment of demonstrated readiness for the target role, compiled from the SkillForge AI readiness, skill-gap, and learning-roadmap engines.

CANDIDATE

Aarav Sharma - B.Tech in Computer Science & Engineering

Undergraduate - Class of 2026 - 14h/week available

TARGET ROLE

AI/ML Engineer

Artificial Intelligence & Data - Demand: Very High - Benchmark: $115,000 / INR 14 - 24 LPA

1. Executive Summary

29 / 100

No change since your last recorded update

Demonstrated Career Readiness

Your demonstrated career readiness for AI/ML Engineer is 29 out of 100. Of this, 16.7 of 50 points comes from weighted core skill coverage, 2.5 of 25 points from passed server-graded assessments, and 10 of 25 points from practical project evidence. You still have 1 critical skill gap to close.

Main strengths

- Verified via assessment: Python Programming (100%)
- 1 portfolio project(s) with submitted repository / live demo links (self-reported)

Most important skill gaps

- Deep Learning with PyTorch (needs Advanced, currently Beginner) - Critical priority
- Docker & Containerization (needs Intermediate, currently None / Novice) - High priority
- FastAPI & REST Model Serving (needs Intermediate, currently None / Novice) - High priority
- Neural Networks & Deep Learning (needs Advanced, currently None / Novice) - High priority
- Classical ML (Scikit-Learn) (needs Advanced, currently Elementary) - High priority

Recommended next action

Take the Linear Algebra & Statistics Skill Assessment to officially verify your proficiency.

2. Readiness Breakdown & Methodology

Core skill coverage	**16.7 / 50**	Role-importance-weighted demonstrated proficiency
Passed assessments	**2.5 / 25**	Passed server-graded technical assessments
Project evidence	**10 / 25**	Completed projects with submitted evidence
Final readiness score	**29 / 100**	Rounded total (clamped 0-100)

Formula and scoring

Readiness = (Core Skill Coverage x 50) + (Passed Assessments x 25) + (Project Evidence x 25), rounded to the nearest whole point and clamped to 0-100. Core skill coverage is the role-importance-weighted fraction of

each required skill demonstrated (capped at the required level). Each passed server-graded assessment contributes toward the assessment component, and completed practical projects with submitted evidence contribute toward the project component.

How missing evidence is handled

Skills with no assessment or project evidence are credited only at the proficiency you self-reported and are labelled self-reported or unverified in this report. Missing evidence lowers the score rather than being assumed, and self-reported skills are never presented as independently verified.

3. Skill Profile by Evidence Source

Tracked skills: 1 assessed, 0 project-backed, 4 self-reported, 5 unverified.

Assessed (server-graded) (1)

Python Programming

Project-backed (verified project) (0)

None recorded.

Self-reported (not independently verified) (4)

SQL & Relational Databases, Linear Algebra & Statistics, Data Wrangling (NumPy & Pandas), Classical ML (Scikit-Learn)

Unverified (5)

Deep Learning with PyTorch, FastAPI & REST Model Serving, LLMs, Prompting & RAG Pipelines, Docker & Containerization, MLOps & Experiment Tracking

Portfolio projects & submitted evidence

- Enterprise Document RAG & QA Pipeline - in progress (no evidence submitted)

4. Prioritized Skill Gaps

1. Deep Learning with PyTorch — Critical

Current: Beginner -> Required: Advanced | Category: Deep Learning
Prerequisites: None | Estimated effort: ~38 hours
Why this ranks here: Required level is 4/5 (currently 1/5). High role importance (92%). All prerequisites satisfied-ready for immediate acceleration.
Next action: Advance to Train a multimodal embedding classifier to match product images with text queries..

2. Docker & Containerization — High

Current: None / Novice -> Required: Intermediate | Category: DevOps & Deployment
Prerequisites: None | Estimated effort: ~20 hours
Why this ranks here: Required level is 3/5 (currently 0/5). High role importance (80%). All prerequisites satisfied-ready for immediate acceleration.
Next action: Begin foundational study using Docker Official Get Started Guide.

3. FastAPI & REST Model Serving — High

Current: None / Novice -> Required: Intermediate | Category: API Development
Prerequisites: None | Estimated effort: ~25 hours
Why this ranks here: Required level is 3/5 (currently 0/5). High role importance (78%). All prerequisites satisfied-ready for immediate acceleration.
Next action: Begin foundational study using FastAPI Official Documentation.

4. Neural Networks & Deep Learning — High

Current: None / Novice -> Required: Advanced | Category: Deep Learning
Prerequisites: Deep Learning with PyTorch | Estimated effort: ~45 hours
Why this ranks here: Required level is 4/5 (currently 0/5). High role importance (90%). Note: Foundational prerequisites [Deep Learning with PyTorch] should be strengthened alongside this.
Next action: Address prerequisite foundations: Deep Learning with PyTorch.

5. Classical ML (Scikit-Learn) High

Current: Elementary -> Required: Advanced | Category: Machine Learning
Prerequisites: None | Estimated effort: ~23 hours
Why this ranks here: Required level is 4/5 (currently 2/5). High role importance (90%). All prerequisites satisfied-ready for immediate acceleration.
Next action: Advance to Build a customer churn prediction engine with probability calibration and explanation..

6. LLMs, Prompting & RAG Pipelines High

Current: None / Novice -> Required: Intermediate | Category: Generative AI
Prerequisites: FastAPI & REST Model Serving | Estimated effort: ~35 hours
Why this ranks here: Required level is 3/5 (currently 0/5). High role importance (88%). Note: Foundational prerequisites [FastAPI & REST Model Serving] should be strengthened alongside this.
Next action: Address prerequisite foundations: FastAPI & REST Model Serving.

7. MLOps & Experiment Tracking High

Current: None / Novice -> Required: Intermediate | Category: DevOps & Deployment
Prerequisites: Docker & Containerization | Estimated effort: ~30 hours
Why this ranks here: Required level is 3/5 (currently 0/5). High role importance (82%). Note: Foundational prerequisites [Docker & Containerization] should be strengthened alongside this.
Next action: Address prerequisite foundations: Docker & Containerization.

8. Python Programming High

Current: Intermediate -> Required: Advanced | Category: Programming Languages
Prerequisites: None | Estimated effort: ~10 hours
Why this ranks here: Required level is 4/5 (currently 3/5). High role importance (95%). All prerequisites satisfied-ready for immediate acceleration.
Next action: Advance to Write a high-performance CLI tool for dataset parsing with multithreading..

9. Data Wrangling (NumPy & Pandas) Medium

Current: Intermediate -> Required: Advanced | Category: Data Engineering
Prerequisites: None | Estimated effort: ~8 hours
Why this ranks here: Required level is 4/5 (currently 3/5). High role importance (90%). All prerequisites satisfied-ready for immediate acceleration.
Next action: Advance to Create an automated exploratory data analysis (EDA) pipeline generating summary reports..

10. Linear Algebra & Statistics Medium

Current: Elementary -> Required: Intermediate | Category: Mathematics
Prerequisites: None | Estimated effort: ~12 hours
Why this ranks here: Required level is 3/5 (currently 2/5). High role importance (85%). All prerequisites satisfied-ready for immediate acceleration.
Next action: Advance to Build a statistical hypothesis testing engine for A/B experiment evaluation..

+ 1 additional lower-priority gaps not shown.

5. Learning Roadmap

Progress: 0 of 11 tasks completed (0%). Estimated hours: ~0 of 254 hrs logged.

Stage 1: Mathematical & Programming Foundations 0/3

Master core Python, linear algebra, vector calculus, and data manipulation.

- [ ] Python Programming (10h)
  Completion criterion: Proficient with list comprehensions, generators, and decorators
- [ ] Data Wrangling (NumPy & Pandas) (8h)
  Completion criterion: Vectorized operations and multidimensional broadcasting in NumPy
- [ ] Linear Algebra & Statistics (12h)
  Completion criterion: Matrix operations, eigenvalues, and SVD

Stage 2: Classical Machine Learning 0/2

Understand feature engineering, supervised/unsupervised algorithms, and evaluation metrics.

- [ ] Classical ML (Scikit-Learn) (23h)
  Completion criterion: Classification, regression, and clustering algorithms
- [ ] SQL & Relational Databases (8h)
  Completion criterion: Complex joins, window functions, and CTEs

Stage 3: Deep Learning & Neural Networks 0/2

Build neural architectures, computer vision, and NLP models with PyTorch.

- [ ] Deep Learning with PyTorch (38h)
  Completion criterion: Tensors, autograd, computational graphs, and GPU acceleration
- [ ] Neural Networks & Deep Learning (45h)
  Completion criterion: Forward and backward propagation, loss functions, and gradient-based optimization

Stage 4: Modern GenAI & LLM Engineering 0/2

Leverage LLMs, prompt engineering, RAG pipelines, and vector databases.

- [ ] FastAPI & REST Model Serving (25h)
  Completion criterion: Asynchronous route handlers and Pydantic validation schemas
- [ ] LLMs, Prompting & RAG Pipelines (35h)
  Completion criterion: Vector embeddings, semantic search, and cosine similarity

Stage 5: MLOps & Production Deployment 0/2

Containerize models, setup CI/CD for ML, and monitor inference endpoints.

- [ ] Docker & Containerization (20h)
  Completion criterion: Writing efficient multi-stage Dockerfiles for Python/ML environments
- [ ] MLOps & Experiment Tracking (30h)
  Completion criterion: Experiment logging with MLflow or Weights & Biases

6. Next Seven Days

Based on your stated availability of 14 hour(s) per week and the priorities above.

3h **Close priority gap: Deep Learning with PyTorch**
Advance to Train a multimodal embedding classifier to match product images with text queries..

2h **Advance project: Enterprise Document RAG & QA Pipeline**
Complete the next milestone: "FastAPI Serving & Streaming".

3h **Close priority gap: Docker & Containerization**
Begin foundational study using Docker Official Get Started Guide.

2h **Advance project: Distributed Token-Bucket Rate Limiter & Telemetry**
Complete the next milestone: "In-Memory Sliding Window Algorithm".

3h **Close priority gap: FastAPI & REST Model Serving**
Begin foundational study using FastAPI Official Documentation.

1h **Close priority gap: Neural Networks & Deep Learning**
Address prerequisite foundations: Deep Learning with PyTorch.

This plan is scoped to the weekly availability you provided.

7. Limitations

What this score measures: the readiness score summarises how well your demonstrated (assessed and project-backed) and self-reported skills currently align with the required skills of the target role. It reflects the evidence SkillForge AI can see in your profile.

What this score does not measure: it does not measure hiring probability, interview performance, communication, teamwork, or market conditions, and it is not a prediction of employment outcomes. Role requirements and market demand are indicative.

Evidence and verification: project evidence (repository and demo links) is self-reported by the student and is

not independently reviewed by SkillForge AI. Assessment results are graded by the SkillForge AI server against fixed answer keys and are not externally proctored or certified.

What your report does not include - your report is based on a complete profile.

This readiness score is an explainable estimate of alignment with curated industry role requirements. It does not guarantee employment or placement.